

Aleena Mary Karatra

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Available: May–December 2026

EDUCATION

Northeastern University, Boston, MA

Khoury College of Computer Sciences

Candidate for a Master of Science in Computer Science

Related courses: Algorithms, Programming Design Paradigms, Machine Learning, Cloud Computing

Fall 2025 – Present

Expected Graduation: May 2027

GPA: 3.50/4.0

Rajagiri School of Engineering and Technology, Kerala, India

Bachelor of Science in Computer Science and Engineering

Related courses: Object Oriented Programming, Data Structures, Algorithms, Machine Learning, AI

2020 – 2024

GPA: 9.05/10.0

TECHNICAL KNOWLEDGE

Languages:	Python, Java, C++, C#, Kotlin, SQL, JavaScript
Tools & Technologies:	Git, Firebase, Arduino, Jupyter, Sigrid
Core Skills:	Debugging, Performance Optimization, Algorithms, Data Structures
ML & Analytics:	Scikit-learn, Pandas, NumPy, NLP, Transfer Learning

WORK EXPERIENCE

Alstom SA, Hyderabad, India

Graduate Engineer Trainee (Software Engineering)

- Improved maintainability of the FBANE rail traffic module by refactoring and modularizing components in a multi-component C++/C# codebase, supporting cross-team development and system reliability.
- Investigated and resolved production issues during NCRTC and Kiwi system deployments, collaborating with engineering and QA teams during commissioning phases.
- Streamlined developer onboarding by leading NCRTC and Kiwi application installations and authoring setup documentation used across the team.
- Developed an MDM communication demo application used for training engineers on system integration workflows and architecture patterns.
- Presented system integration workflows and installation processes to internal engineering teams.

July 2024 – July 2025

PROJECTS

Emotional Text-to-Speech Synthesizer (Multilingual NLP)

- Built a multilingual sentiment-aware TTS pipeline using BERT emotion classification and StarGAN voice synthesis to support low-resource languages
- Evaluated emotional prosody injection by analyzing acoustic features including amplitude and frequency shifts

Oct 2023 – May 2024

Recipe Recommendation System - IEEE Hackathon Winner

- Won 1st place at an Intel-sponsored IEEE hackathon (INR 30,000 prize).
- Implemented collaborative filtering and content-based ranking for personalized recommendations.
- Built a real-time recipe recommendation engine in Python using Scikit-learn.

May 2023

Odor Detection and Notification System (IoT + Automation)

- Built an end-to-end IoT solution for campus restroom maintenance using ammonia sensors to detect odor, trigger automatic air freshener deployment, and send recurring mobile alerts to cleaning staff
- Designed unit-level tests and documented system behavior for maintainability

Apr 2023 – Sep 2023

INTERESTS/ACTIVITIES

- Chair, IEEE Women in Engineering (2023–24) — Led technical workshops, managed chapter initiatives, and mentored peers in engineering and research activities.
- Secretary General & Chair, Model United Nations — Directed large-scale academic conferences and developed strong public-speaking and leadership skills.